

Equity Leverage Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Leverage Sensitivity (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by ?. A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the ? five-factor model plus a momentum factor of 18 (15) bps/month with a t-statistic of 2.30 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.37.

1 Introduction

The following automatically generated report tests the asset pricing implications of Equity Leverage Sensitivity (ELS), and its robustness in predicting returns in the cross-section of equities. It is produced using the methodology of ?, from input data consisting of firm-month observations for the proposed predictor.¹

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different fiscal year conventions. construction of our Equity Leverage Sensitivity signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares rather than their market value. Debt in current liabilities (DLC) captures the portion of long-term debt that is due within one year, including the current portion of long-term debt, sinking fund requirements, and other long-term debt obligations reclassified as current. This variable represents near-term debt obligations that firms must satisfy within the current operating cycle. construct the Equity Leverage Sensitivity signal by calculating the year-over-year change in common stock scaled by lagged short-term debt obligations. Specifically, we compute $(CSTK(t) - CSTK(t-1)) / DLC(t-1)$, where t denotes the current fiscal year. This ratio captures the relationship between changes in equity capital structure and existing short-term debt obligations, providing insight into how

¹It used version v0.4.1 of the publicly available code repository at <https://github.com/velikov-mihail/AssayingAnomalies>. See more details at <http://AssayingAnomalies.com>.

firms adjust their equity base relative to their near-term debt burden. The numerator measures the annual change in the book value of common equity, which can result from new equity issuances, stock repurchases, or changes in par value. By scaling this change by the prior year’s short-term debt obligations, we normalize for firm size and debt structure differences across companies. We require non-missing values for both CSTK and DLC in consecutive years to compute the signal, and we exclude observations where $\text{DLC}(t-1)$ equals zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is -1.46 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal’s interquartile range spans -0.52 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 5.22% of CRSP names, which on average make up 6.95% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the ? three-factor model (FF3) and its variation that adds momentum (FF4), the ? five-factor model (FF5), and its variation that

adds momentum factor used in ? (FF6). The table shows that the long/short ELS strategy earns an average return of 0.26% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.17 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the ? six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.05 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least \$1,347 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of α . These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from α . The net average returns reported in the first column range between 18-32bps/month. The lowest return, (18 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ELS strategy achieves an average return of 24 bps/month with a t-statistic of 2.67. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 16 to 26 bps/month with t-statistics between 1.80 and 2.90.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.² The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy’s gross (net) Sharpe ratio of 0.42 (0.36) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).³ Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$4.63 which ranks the ELS strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$3.27 which ranks the ELS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and ? net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 54 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the

²The anomalies come from March, 2022 release of the ? open source asset pricing dataset.

³The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in ?, that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 72 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 209 filtered anomaly signals.⁴ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns

⁴When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 3.51, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on ELS is 2.51.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.52, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.37.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies

that combine multiple anomalies following ?. The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELS signal.⁵ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ELS grows to \$2284.79.

8 Conclusion

This study provides compelling evidence that Equity Leverage Sensitivity (ELS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.36, which remains economically meaningful after accounting for transaction costs. Most notably, the strategy’s ability to generate a statistically significant alpha of 19 basis points per month (t-statistic = 2.37) relative to the Fama-French five factors, momentum, and six closely related financing-based strategies underscores its unique information content beyond existing predictors in the factor zoo.

⁵We filter the 207 ? anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELS is available.

The practical implications of these findings are significant for both academic researchers and investment practitioners. The persistence of the ELS effect after controlling for related equity financing variables suggests that markets may not fully incorporate the information contained in firms' leverage sensitivity into asset prices. This inefficiency presents potential opportunities for sophisticated investors while contributing to our understanding of how capital structure dynamics influence expected returns. The strategy's performance after transaction costs indicates that the anomaly is not merely a statistical artifact but represents a potentially exploitable market inefficiency.

However, several limitations warrant consideration. First, while our analysis controls for numerous factors, the possibility of omitted risk factors that could explain the ELS premium cannot be entirely ruled out. Second, the implementation of ELS-based strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, the stability of the ELS effect across different market regimes and its performance during extreme market conditions require further investigation.

Future research should explore several promising avenues. Examining the economic mechanisms underlying the ELS effect could provide deeper insights into why this signal predicts returns. International evidence would help establish whether the phenomenon is U.S.-specific or represents a global asset pricing regularity. Additionally, investigating the interaction between ELS and firm characteristics such as financial constraints, growth opportunities, and corporate governance could enhance our understanding of when and why the signal is most effective. Finally, exploring the time-varying nature of the ELS premium and its relationship to macroeconomic conditions could yield valuable insights for dynamic portfolio construction and risk management strategies.

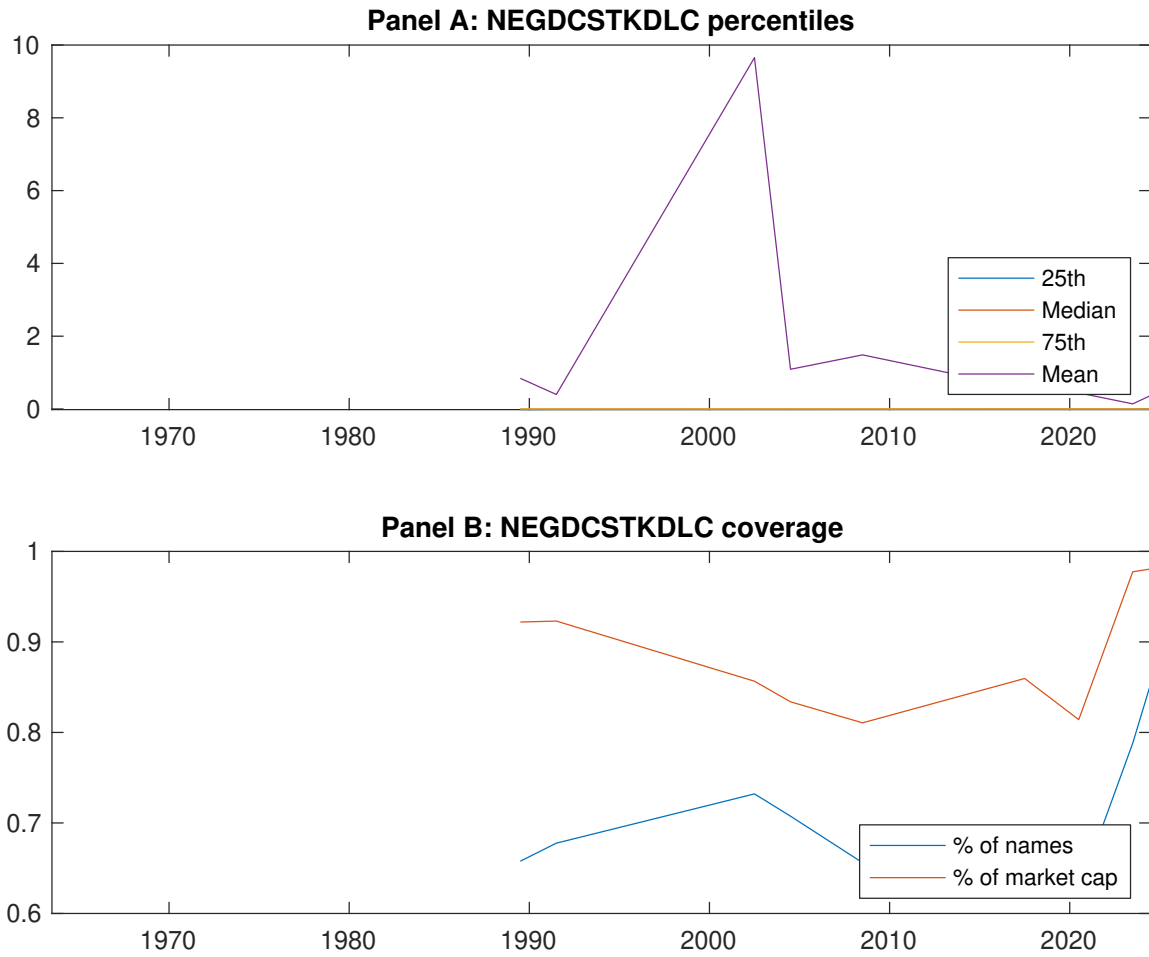


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, ? three-factor model, ? three-factor model augmented with the ? momentum factor, ? five-factor model, and the ? five-factor model augmented with the ? momentum factor following ?. Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the ? five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.89]	0.50 [2.85]	0.58 [3.35]	0.62 [3.88]	0.73 [4.59]	0.26 [3.33]
α_{CAPM}	-0.08 [-1.73]	-0.10 [-2.28]	-0.01 [-0.24]	0.08 [1.58]	0.20 [3.72]	0.28 [3.57]
α_{FF3}	-0.08 [-1.67]	-0.09 [-1.98]	-0.05 [-1.02]	0.02 [0.37]	0.15 [3.00]	0.23 [3.00]
α_{FF4}	-0.09 [-1.74]	-0.08 [-1.71]	-0.04 [-0.76]	0.02 [0.36]	0.15 [2.97]	0.24 [3.02]
α_{FF5}	-0.11 [-2.28]	-0.06 [-1.28]	-0.11 [-2.23]	-0.07 [-1.75]	0.06 [1.14]	0.17 [2.17]
α_{FF6}	-0.12 [-2.30]	-0.05 [-1.14]	-0.09 [-1.90]	-0.07 [-1.53]	0.07 [1.32]	0.18 [2.30]
Panel B: ? 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	0.96 [78.72]	1.02 [89.17]	1.03 [90.34]	1.00 [96.64]	0.98 [82.46]	0.02 [0.94]
β_{SMB}	0.00 [0.27]	-0.02 [-0.96]	0.00 [0.18]	-0.09 [-5.71]	-0.04 [-2.28]	-0.04 [-1.59]
β_{HML}	0.00 [0.11]	0.00 [0.06]	0.05 [2.46]	0.10 [4.89]	0.05 [2.10]	0.05 [1.24]
β_{RMW}	0.10 [4.10]	-0.04 [-1.78]	0.12 [5.50]	0.14 [6.71]	0.16 [6.94]	0.06 [1.70]
β_{CMA}	-0.01 [-0.39]	-0.09 [-2.63]	0.09 [2.87]	0.23 [7.88]	0.21 [6.10]	0.22 [4.05]
β_{UMD}	0.00 [0.34]	-0.01 [-0.74]	-0.02 [-1.84]	-0.01 [-1.21]	-0.01 [-1.20]	-0.02 [-0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	583	564	501	562	608	
me (\$10 ⁶)	1420	1347	1902	2046	2308	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, 3 three-factor model, 3 three-factor model augmented with the 3 momentum factor, 3 five-factor model, and the 3 five-factor model augmented with the 3 momentum factor following 3. Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and 3 generalized alphas as prescribed by 3. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.33]	0.28 [3.57]	0.23 [3.00]	0.24 [3.02]	0.17 [2.17]	0.18 [2.30]
Quintile	NYSE	EW	0.51 [7.59]	0.58 [8.95]	0.49 [8.26]	0.42 [7.04]	0.36 [6.26]	0.30 [5.37]
Quintile	Name	VW	0.29 [3.78]	0.31 [4.08]	0.26 [3.51]	0.26 [3.46]	0.20 [2.60]	0.20 [2.68]
Quintile	Cap	VW	0.22 [2.77]	0.26 [3.26]	0.20 [2.65]	0.20 [2.60]	0.13 [1.71]	0.14 [1.80]
Decile	NYSE	VW	0.36 [3.98]	0.41 [4.58]	0.33 [3.82]	0.34 [3.87]	0.24 [2.70]	0.26 [2.89]
Panel B: Net Returns and 3 generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [2.85]	0.24 [3.06]	0.19 [2.48]	0.19 [2.51]	0.14 [1.86]	0.15 [1.94]
Quintile	NYSE	EW	0.32 [4.39]	0.41 [5.69]	0.32 [4.87]	0.28 [4.32]	0.18 [2.97]	0.16 [2.65]
Quintile	Name	VW	0.25 [3.29]	0.27 [3.53]	0.22 [2.96]	0.22 [2.95]	0.17 [2.31]	0.18 [2.36]
Quintile	Cap	VW	0.18 [2.31]	0.22 [2.81]	0.17 [2.19]	0.17 [2.17]	0.12 [1.54]	0.12 [1.59]
Decile	NYSE	VW	0.32 [3.51]	0.36 [3.92]	0.28 [3.19]	0.28 [3.25]	0.20 [2.31]	0.21 [2.45]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ELS Quintiles					ELS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.51 [1.98]	0.74 [2.92]	0.82 [3.27]	0.89 [3.71]	1.03 [4.46]	0.55 [5.62]	0.61 [6.34]	0.54 [5.83]	0.48 [5.15]	0.38 [4.25]	0.35 [3.80]
	(2)	0.64 [2.76]	0.73 [3.11]	0.76 [3.28]	0.88 [3.85]	0.97 [4.54]	0.33 [3.50]	0.40 [4.31]	0.30 [3.36]	0.27 [3.00]	0.24 [2.63]	0.22 [2.40]
	(3)	0.54 [2.66]	0.57 [2.66]	0.82 [3.72]	0.77 [3.76]	0.96 [4.91]	0.42 [5.04]	0.45 [5.39]	0.36 [4.58]	0.31 [3.87]	0.30 [3.74]	0.26 [3.22]
	(4)	0.57 [2.93]	0.62 [3.17]	0.76 [3.84]	0.75 [3.93]	0.78 [4.29]	0.22 [2.43]	0.26 [2.95]	0.16 [1.98]	0.16 [1.98]	-0.03 [-0.35]	-0.01 [-0.10]
	(5)	0.44 [2.67]	0.50 [2.91]	0.45 [2.68]	0.57 [3.61]	0.68 [4.28]	0.24 [2.67]	0.26 [2.90]	0.21 [2.30]	0.23 [2.54]	0.16 [1.80]	0.19 [2.09]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ELS Quintiles					ELS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	308	306	305	305	300	26	26	27	22	21	
	(2)	88	87	88	88	87	44	43	44	43	43	
	(3)	66	65	66	65	65	78	77	80	81	80	
	(4)	56	56	56	57	57	170	173	181	177	183	
(5)	54	55	54	54	54	1220	1367	1568	1491	1765		

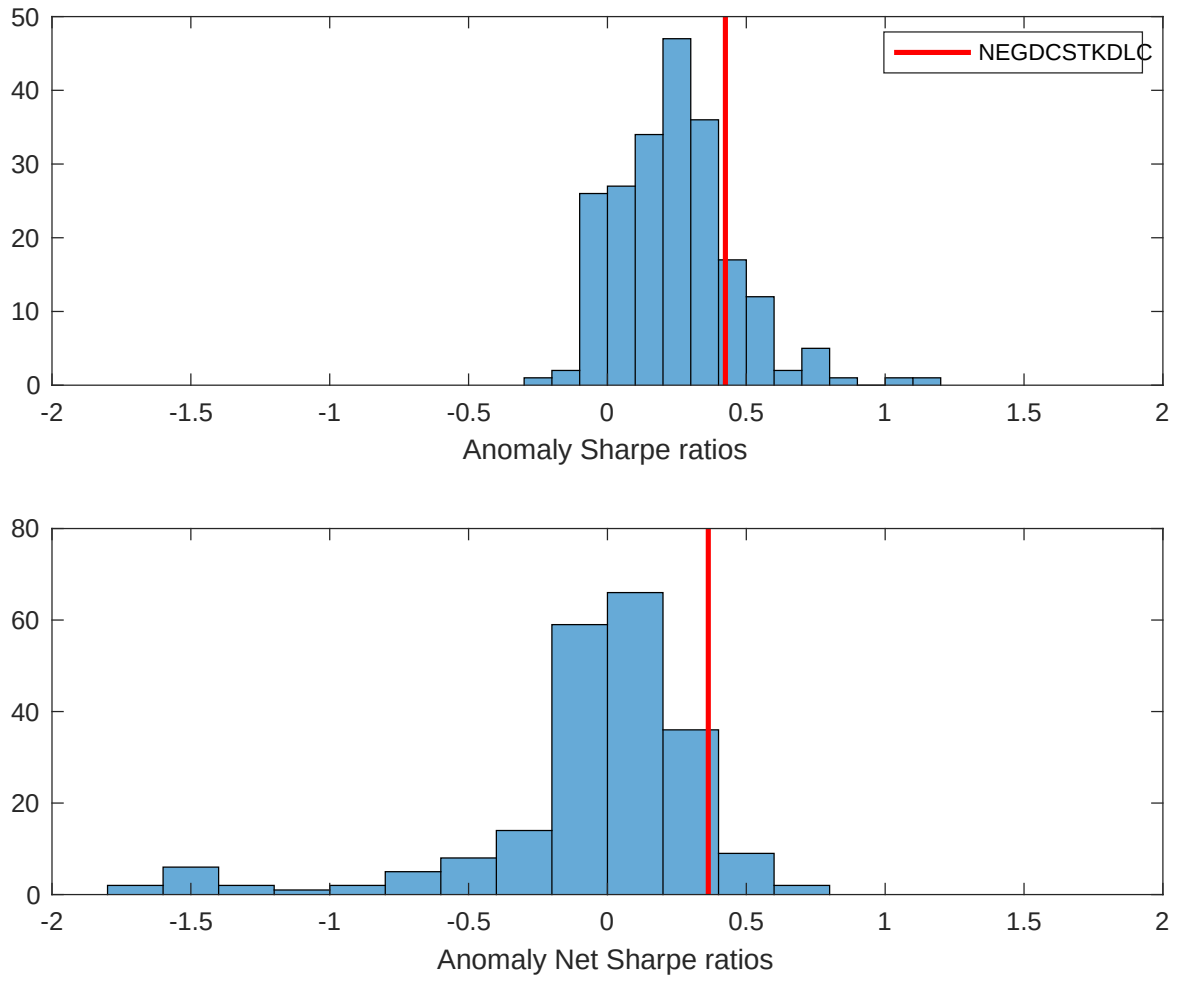


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

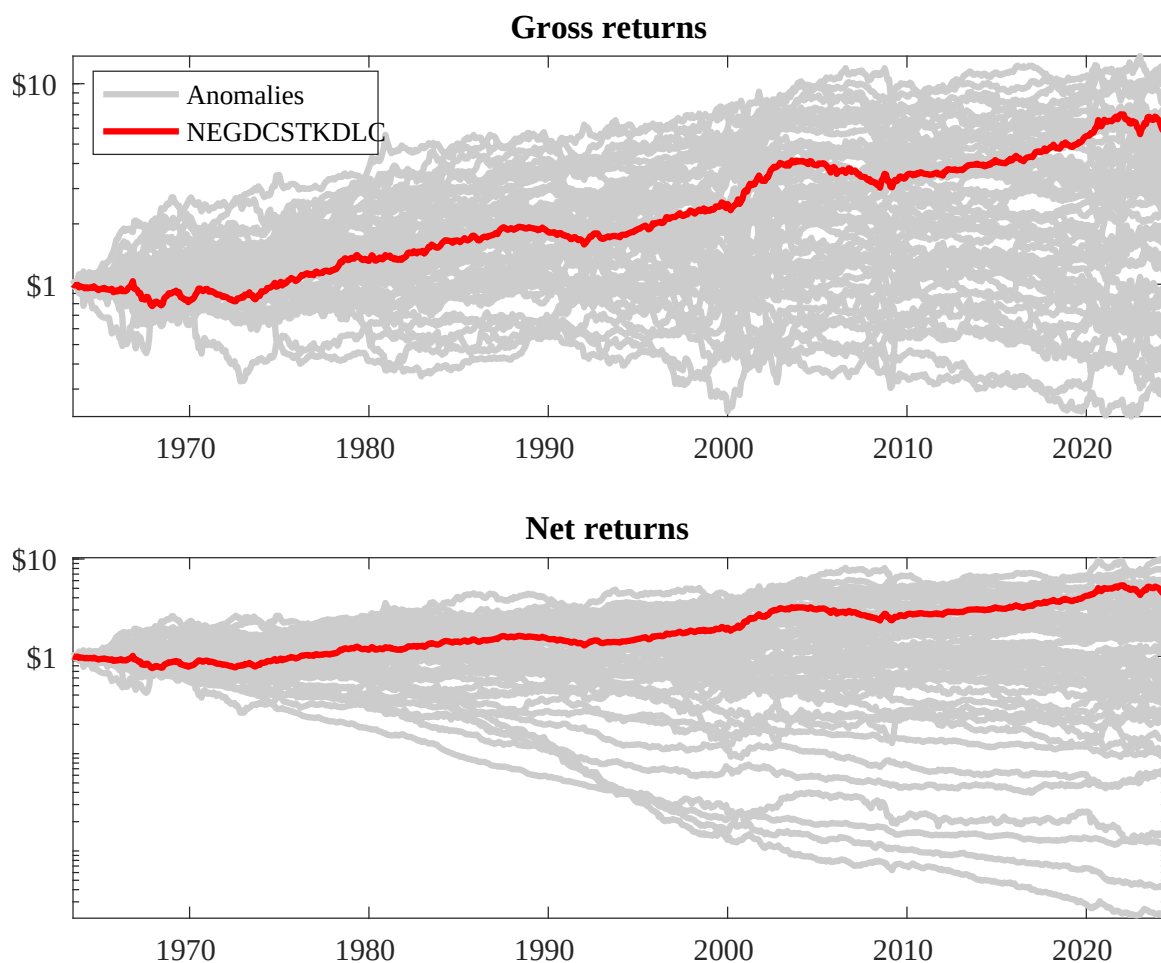
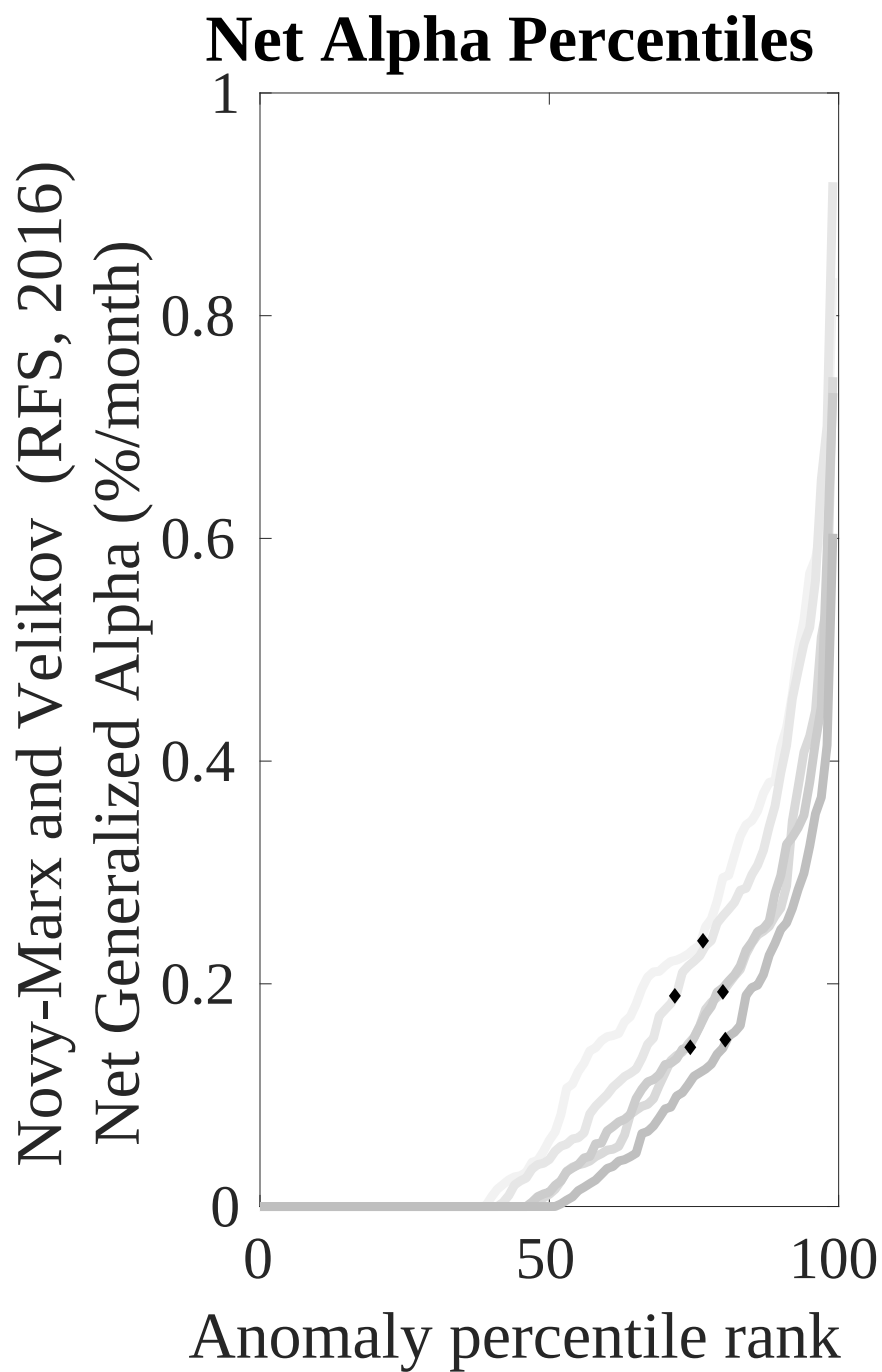
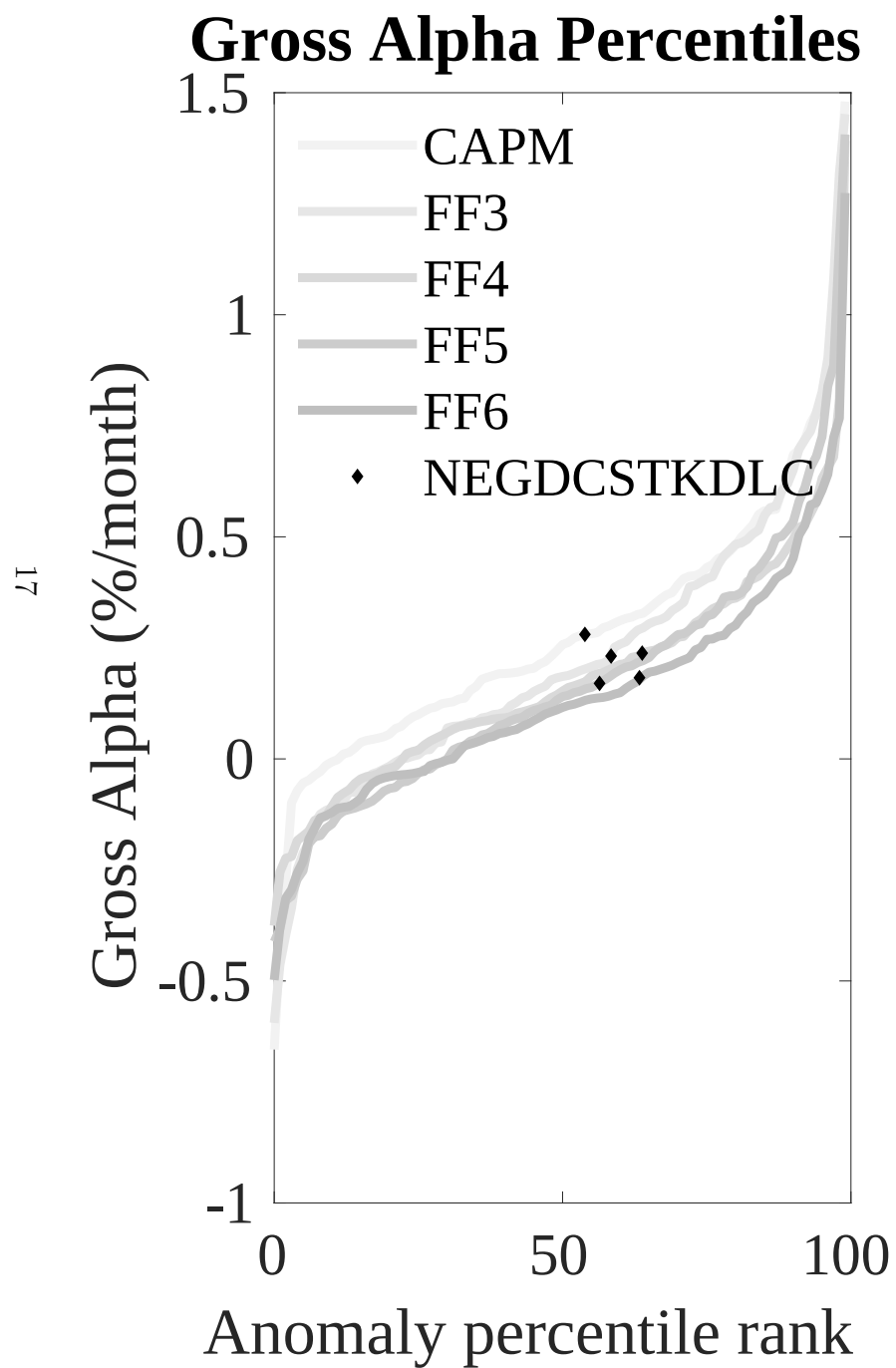


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



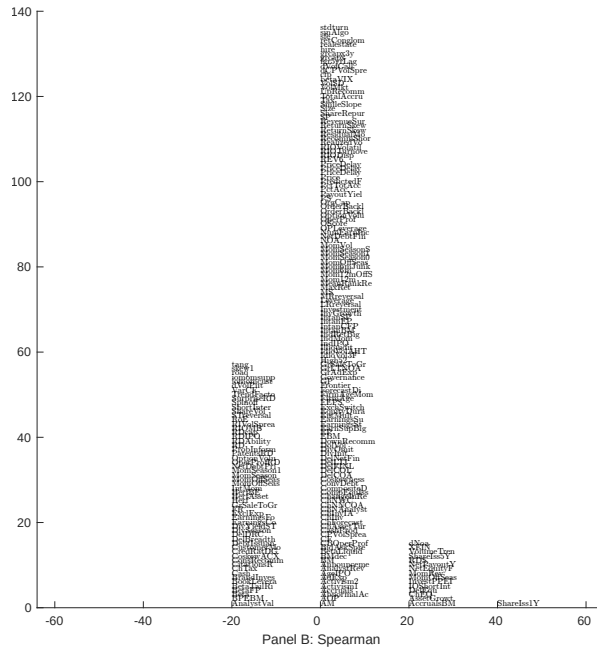
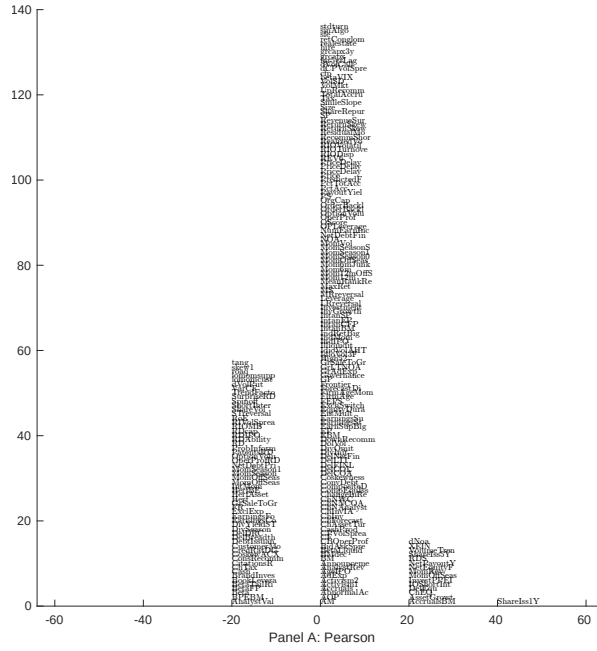


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

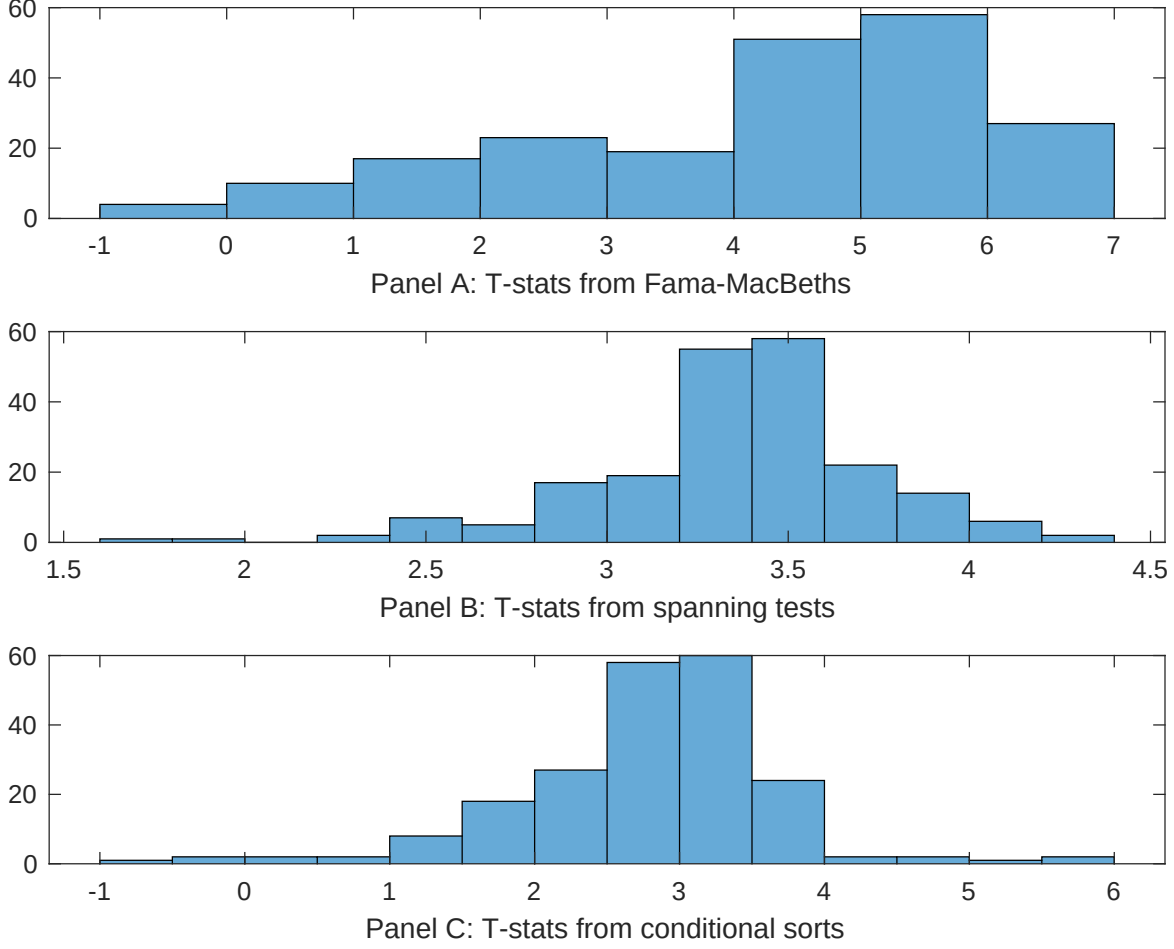


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.01]	0.12 [5.34]	0.17 [6.93]	0.13 [6.44]	0.12 [5.67]	0.13 [5.31]	0.13 [4.91]
ELS	0.68 [4.82]	0.69 [4.60]	0.64 [3.88]	0.70 [4.70]	0.68 [4.19]	0.78 [3.51]	0.42 [2.51]
Anomaly 1	0.15 [6.33]						0.65 [2.73]
Anomaly 2		0.20 [1.69]					0.11 [2.24]
Anomaly 3			0.43 [3.50]				-0.94 [-0.65]
Anomaly 4				0.30 [5.58]			0.30 [6.41]
Anomaly 5					0.13 [3.26]		0.12 [2.02]
Anomaly 6						0.18 [2.91]	-0.14 [-1.59]
# months	715	704	720	715	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the ? five-factor model augmented with the ? momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [2.16]	0.21 [2.70]	0.16 [2.13]	0.12 [1.52]	0.18 [2.27]	0.23 [2.75]	0.19 [2.37]
Anomaly 1	22.60 [5.75]						2.63 [0.49]
Anomaly 2		17.08 [5.68]					11.95 [2.92]
Anomaly 3			26.81 [6.42]				30.44 [4.66]
Anomaly 4				24.86 [6.01]			16.67 [3.14]
Anomaly 5					12.95 [3.32]		-10.45 [-1.74]
Anomaly 6						7.33 [1.92]	-12.09 [-2.51]
mkt	2.57 [1.39]	4.65 [2.44]	2.87 [1.55]	3.12 [1.69]	1.75 [0.93]	2.82 [1.36]	3.55 [1.74]
smb	-2.46 [-0.92]	-0.54 [-0.20]	-4.15 [-1.56]	-1.68 [-0.63]	-3.41 [-1.26]	1.87 [0.57]	-3.55 [-1.09]
hml	5.29 [1.48]	-0.83 [-0.22]	2.50 [0.70]	7.85 [2.20]	4.09 [1.12]	6.08 [1.61]	2.55 [0.66]
rmw	-5.89 [-1.41]	-3.62 [-0.90]	7.54 [2.10]	-2.57 [-0.66]	7.46 [2.03]	7.61 [1.69]	4.21 [0.86]
cma	6.95 [1.21]	6.67 [1.15]	-4.82 [-0.73]	8.96 [1.60]	8.11 [1.23]	6.81 [1.11]	-19.19 [-2.64]
umd	-2.67 [-1.45]	-0.28 [-0.15]	-2.10 [-1.15]	-3.26 [-1.77]	-1.31 [-0.70]	-1.38 [-0.71]	-2.20 [-1.15]
# months	728	728	732	728	732	630	626
$\bar{R}^2(\%)$	11	11	12	12	8	6	14

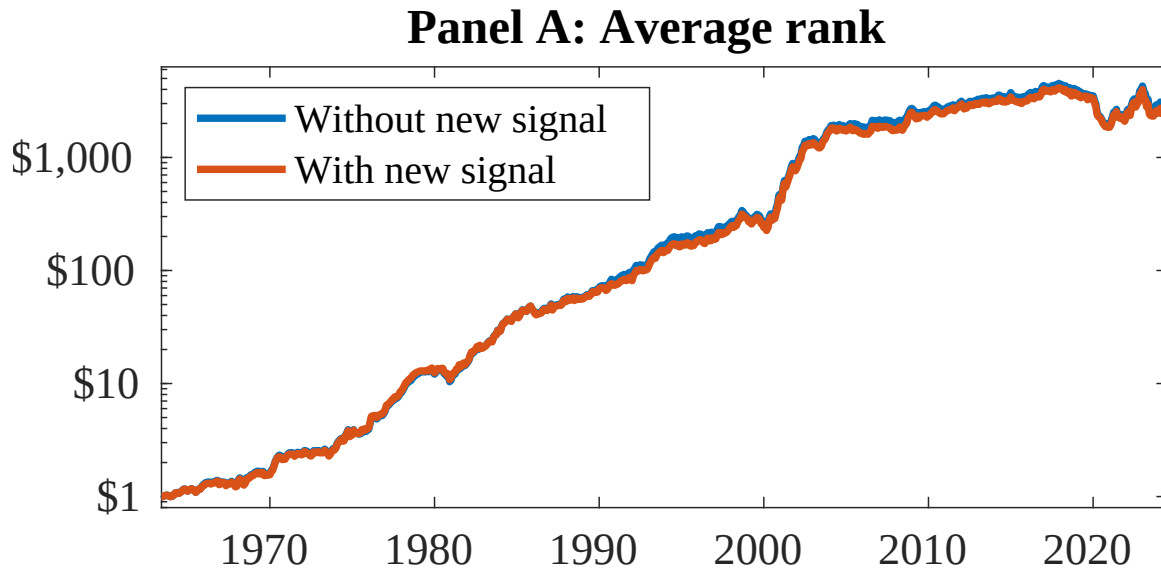


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following ?. In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.